Project III Plan
Image Generation using Generative Models

Michał Bober
Maciej Karaśkiewicz

Warsaw University of Technology — Deep Learning 2026

1 Problem Statement

The goal of this project is to investigate generative models for image generation using the Cat Dataset [1]. The task is to train neural networks capable of generating realistic images of cats.

Image generation is a challenging problem because the model must learn not only low-level visual features such as colours, edges, and textures, but also higher-level structure such as the shape of the cat's head, eyes, ears, and overall pose. The dataset contains cat images with different resolutions and visual conditions, therefore an important part of the project will be to design a suitable preprocessing pipeline.

The main objectives of this project are:

- preprocess the Cat Dataset and convert all images to a fixed resolution,
- implement and compare selected generative models,
- analyse the influence of selected hyperparameters on generated image quality,
- evaluate generated images quantitatively using Fréchet Inception Distance (FID),
- assess generated images qualitatively,
- analyse the mode collapse problem if it occurs,
- perform latent space interpolation between two generated samples.

2 Planned Experiments

Experiments will be repeated multiple times where computationally feasible. The generated images will be evaluated using both quantitative metrics and visual inspection. Due to limited computing resources, the models will be trained on resized images and, if necessary, on a reduced subset of the dataset.

2.1 Dataset Preprocessing

The Cat Dataset contains images of cats with different resolutions. Since neural networks require fixed-size inputs, all images will be preprocessed before training.

Planned preprocessing steps:

- load all available cat images from the dataset,
- remove corrupted or unreadable files,

- resize images to a fixed resolution,
- normalise pixel values to the range required by the selected model,

We plan to test the following image resolutions:

Resolution	Purpose
64×64	faster training and baseline experiments
128×128	improved visual quality if compute allows

Table 1: Planned image resolutions

2.2 Architecture Comparison

We plan to compare several generative approaches. The final choice may be adjusted depending on computational cost and training stability.

Model	Architecture characteristics
DCGAN	Convolutional generator and discriminator
VAE	Encoder-decoder model with continuous latent space

Table 2: Planned generative models for comparison

The DCGAN model will be used as a relatively simple baseline. The VAE model will allow us to analyse a structured latent space and perform interpolation between latent vectors.

2.3 Hyperparameter Experiments

The influence of selected hyperparameters will be analysed. Since training generative models is computationally expensive, we will focus on a limited but meaningful set of configurations.

2.3.1 Training Hyperparameters

Hyperparameter	Values to test
learning rate	10^{-4}, $2 \cdot 10^{-4}$, 10^{-3}
batch size	32, 64

Table 3: Training hyperparameters to investigate

2.3.2 Model-Specific Hyperparameters

Hyperparameter	Values to test
latent dimension	64, 128, 256
generator feature maps	64, 128
discriminator feature maps	64, 128

Table 4: Model-specific hyperparameters to investigate

2.4 Evaluation

The generated images will be evaluated quantitatively and qualitatively.

2.4.1 Quantitative Evaluation

We will use Fréchet Inception Distance (FID) to compare the distribution of generated images with the distribution of real images [6]. Lower FID indicates that generated samples are closer to real images in the feature space of an Inception network.

For each selected model, we will generate a fixed number of samples and compute FID against a subset of real images from the Cat Dataset.

2.4.2 Qualitative Evaluation

In addition to FID, generated samples will be visually inspected. The qualitative analysis will focus on:

- overall realism of generated images,
- sharpness and texture quality,
- presence of cat-like facial structure,
- diversity of generated samples,
- visible artifacts or repeated patterns.

2.5 Mode Collapse Analysis

If mode collapse occurs, especially in GAN-based models, we will analyse it by inspecting generated samples and measuring diversity between outputs. Mode collapse will be considered present if the model repeatedly generates very similar images despite different latent vectors.

Possible mitigation strategies include:

- changing learning rate,
- changing discriminator and generator capacity,
- adding label smoothing,
- using different GAN loss variants,
- training for more epochs or with a smaller learning rate.

2.6 Latent Space Interpolation

We will select two generated images together with their latent vectors. Then, we will linearly interpolate between these two latent vectors and generate intermediate images.

Given two latent vectors z_1 and z_2, interpolation will be performed as:

$$z_\alpha = (1 - \alpha)z_1 + \alpha z_2$$

where α will take 10 equally spaced values between 0 and 1. This will produce 10 images in total: the two original generated images and 8 intermediate samples.

The goal of this experiment is to analyse whether the model learns a meaningful latent space in which small changes in the latent vector correspond to smooth visual changes in generated images.

3 Division of Work

Task	Owner
Dataset preprocessing pipeline	Maciej Karaśkiewicz
DCGAN implementation	Michał Bober
VAE implementation	Maciej Karaśkiewicz
FID evaluation pipeline	Maciej Karaśkiewicz
Latent interpolation experiment	Michał Bober
Hyperparameter experiments	Both
Results analysis	Both
Writing the report	Both
Presentation slides	Both

Table 5: Planned division of work

4 Project Timeline

Date	Milestone
May 19	Plan submitted
May 26	Dataset preprocessing and baseline DCGAN ready
Jun. 2	VAE and evaluation pipeline ready
Jun. 9	Running experiments, FID evaluation, latent interpolation
Jun. 16	Deadline — all artifacts uploaded to LeON

Table 6: Project timeline

5 Use of LLMs in the Process of Preparing the Plan

Large Language Models were used during the preparation of this plan to assist with literature search and checking grammar. The final experimental decisions and project implementation will be verified and carried out by the authors.

References

[1] Chris Crawford. *Cat Dataset.* Kaggle. `https://www.kaggle.com/datasets/crawford/cat-dataset`

[2] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. *Generative Adversarial Nets.* Advances in Neural Information Processing Systems, 2014.

[3] Alec Radford, Luke Metz, and Soumith Chintala. *Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks.* arXiv preprint arXiv:1511.06434, 2015.

[4] Diederik P. Kingma and Max Welling. *Auto-Encoding Variational Bayes.* arXiv preprint arXiv:1312.6114, 2013.

[5] Jonathan Ho, Ajay Jain, and Pieter Abbeel. *Denoising Diffusion Probabilistic Models.* Advances in Neural Information Processing Systems, 2020.

[6] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. *GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium.* Advances in Neural Information Processing Systems, 2017.